

### Question Bank – Matter Around Us (100 MCQs)

1. Matter is anything that has:

- A) Shape and colour
- B) Mass and occupies space
- C) Taste and smell
- D) Length and height

**Answer: B**

2. Which of the following is *not* matter?

- A) Air
- B) Water
- C) Stone
- D) Love

**Answer: D**

3. The smallest unit of matter is:

- A) Molecule
- B) Atom
- C) Neutron
- D) Proton

**Answer: B**

4. The air you breathe is an example of:

- A) Energy
- B) Matter
- C) Emotion
- D) Force

**Answer: B**

5. Which instrument is used to view the structure of atoms?

- A) Telescope
- B) SEM and TEM
- C) Periscope
- D) Microscope

**Answer: B**

6. An atom consists of:

- A) Only protons
- B) Protons, neutrons, electrons
- C) Only electrons
- D) Protons and electrons only

**Answer: B**

7. Neutrons and protons are found in:

- A) Orbitals
- B) Nucleus
- C) Shells
- D) Outside atom

**Answer: B**

8. Electrons are located:

- A) In the nucleus
- B) Around the nucleus
- C) In protons
- D) Inside neutrons

**Answer: B**

9. A single drop of water contains about:

- A)  $10^3$  particles
- B)  $10^6$  particles
- C)  $10^{21}$  particles
- D)  $10^{24}$  particles

**Answer: C**

10. A dot made by a pen has more than:

- A) 200 particles
- B) 2000 particles
- C) 2 lakh particles
- D) 20 crore particles

**Answer: C**

11. Particles of matter are:

- A) Big
- B) Very small
- C) Visible easily
- D) Immovable

**Answer: B**

12. The force of attraction between particles is:

- A) Same in all matter
- B) Different in different states
- C) Always maximum
- D) Always zero

**Answer: B**

13. Particles of matter are:

- A) Stationary
- B) Constantly moving
- C) Still at high temperature
- D) Moving only in solids

**Answer: B**

14. Space between particles in matter is:

- A) Same for all
- B) Different for solids, liquids, gases
- C) Not present
- D) Infinite

**Answer: B**

15. How many physical states of matter are there?

- A) 2
- B) 3
- C) 4
- D) 5

**Answer: B**

16. Particles in solids are:

- A) Tightly packed
- B) Loosely packed
- C) Randomly arranged
- D) Far apart

**Answer: A**

17. Solids have:

- A) No definite shape
- B) No definite volume
- C) Both definite shape and volume
- D) Neither shape nor volume

**Answer: C**

18. Liquids have:

- A) Definite shape but no volume
- B) Definite volume but no shape
- C) No shape, no volume
- D) Definite shape and volume

**Answer: B**

19. Gases have:

- A) Definite shape only
- B) Definite volume only
- C) Definite shape and volume
- D) No definite shape and volume

**Answer: D**

20. Example of solid is:

- A) Air
- B) Water
- C) Stone
- D) Oxygen

**Answer: C**

21. Diffusion is movement of particles from:

- A) Lower to higher concentration
- B) Higher to lower concentration
- C) Solid to gas
- D) One state to another

**Answer: B**

22. Diffusion takes place in:

- A) Solids only
- B) Liquids and gases
- C) Gases only
- D) None

**Answer: B**

23. Compressibility is:

- A) Ability to expand
- B) Ability to reduce volume under pressure
- C) Ability to diffuse
- D) Ability to dissolve

**Answer: B**

24. Which state is highly compressible?

- A) Solids
- B) Liquids
- C) Gases
- D) All equal

**Answer: C**

25. Chalk powder in a syringe cannot be compressed because:

- A) It has space
- B) Particles are tightly packed
- C) Particles are moving
- D) It is liquid

**Answer: B**

26. Which state has maximum diffusion?

- A) Solid
- B) Liquid
- C) Gas
- D) None

**Answer: C**

27. Force of attraction between particles is maximum in:

- A) Solids
- B) Liquids
- C) Gases
- D) Plasma

**Answer: A**

28. Movement of particles is highest in:

- A) Solid
- B) Liquid
- C) Gas
- D) Ice

**Answer: C**

29. Liquids are:

- A) Not compressible

- B) Highly compressible
- C) Hardly compressible
- D) Infinite compressible

**Answer: C**

30. Solids have a:

- A) Definite shape only
- B) Definite volume only
- C) Both definite shape and volume
- D) Neither

**Answer: C**

31. Liquefaction is conversion of:

- A) Liquid to gas
- B) Solid to liquid
- C) Gas to liquid
- D) Solid to gas

**Answer: C**

32. To convert gas into liquid, we:

- A) Increase temperature
- B) Decrease pressure
- C) Apply pressure and reduce temperature
- D) Keep in sunlight

**Answer: C**

33. The process of solid changing into liquid by heating is:

- A) Freezing
- B) Melting
- C) Condensation
- D) Vaporization

**Answer: B**

34. Conversion of liquid to gas is:

- A) Melting
- B) Evaporation
- C) Condensation
- D) Freezing

**Answer: B**

35. Reverse of evaporation is:

- A) Melting
- B) Condensation
- C) Liquefaction
- D) Sublimation

**Answer: B**

36. Pure substances consist of:

- A) One kind of particles
- B) Two kinds of particles
- C) Many kinds of particles
- D) Only liquids

**Answer: A**

36. Pure substances are classified into:

- A) Compounds only
- B) Elements and compounds
- C) Solids and gases
- D) Liquids and solids

**Answer: B**

37. Gold, copper, silver are examples of:

- A) Compounds
- B) Elements
- C) Mixtures
- D) Non-metals

**Answer: B**

38. Water (H<sub>2</sub>O) is a:

- A) Mixture
- B) Element
- C) Compound
- D) Alloy

**Answer: C**

39. Salt (NaCl) is a:

- A) Compound
- B) Mixture
- C) Element
- D) Alloy

**Answer: A**

40. Mixture is formed by:

- A) Chemical combination
- B) Physical combination
- C) Both chemical and physical
- D) None

**Answer: B**

41. 22-carat gold is:

- A) Pure gold
- B) Mixture of gold with other metals
- C) Compound of gold
- D) Impure silver

**Answer: B**

42. Air is an example of:

- A) Pure substance
- B) Compound
- C) Mixture
- D) Alloy

**Answer: C**

43. 24k gold contains:

- A) 91.6% gold
- B) 99.9% gold
- C) 75% gold
- D) 50% gold

**Answer: B**

44. The purity of gold is measured in:

- A) Grams
- B) Litres
- C) Carat (k)
- D) Mole

**Answer: C**

### **Separation of Mixtures**

46. Separation is needed to:

- A) Get pure substances
- B) Remove harmful impurities
- C) Get useful components
- D) All of the above

**Answer: D**

47. Removing stones from rice is:

- A) Sieving
- B) Winnowing
- C) Handpicking
- D) Churning

**Answer: C**

48. Tea leaves from tea are separated by:

- A) Sedimentation
- B) Filtration
- C) Decantation
- D) Winnowing

**Answer: B**

49. Sedimentation means:

- A) Solids float
- B) Solids settle at bottom
- C) Liquids separate
- D) Gas diffuses

**Answer: B**

50. Transferring supernatant liquid without disturbing sediments is:

- A) Filtration
- B) Churning
- C) Decantation
- D) Evaporation

**Answer: C**

51. Separating butter from curd is:

- A) Winnowing
- B) Filtration
- C) Churning
- D) Sieving

**Answer: C**

52. Sieving separates:

- A) Different sizes of solids
- B) Solids and liquids
- C) Liquids only
- D) Solids and gases

**Answer: A**

53. Winnowing is done to separate:

- A) Heavy and light solids
- B) Solid and liquid
- C) Solid and gas
- D) Liquids

**Answer: A**

54. Separation of grains from stalks is:

- A) Handpicking
- B) Threshing
- C) Winnowing
- D) Sieving

**Answer: B**

55. Magnetic separation separates substances based on:

- A) Size
- B) Color
- C) Magnetic property
- D) Shape

**Answer: C**

56. An example of magnetic substance is:

- A) Iron
- B) Gold
- C) Silver
- D) Copper

**Answer: A**

57. Evaporation converts:

- A) Solid to gas
- B) Liquid to vapor
- C) Gas to liquid
- D) Liquid to solid

**Answer: B**

58. Condensation converts:

- A) Vapor to liquid
- B) Liquid to vapor
- C) Solid to gas
- D) Liquid to solid

**Answer: A**

59. Adulteration means mixing:

- A) Pure substances only
- B) Harmful and unwanted substances
- C) Sweet substances
- D) Clean substances

**Answer: B**

60. Adulteration is harmful because:

- A) It looks good
- B) It causes health hazard
- C) It makes food tasty
- D) It improves nutrition

**Answer: B**

61. Used tea leaves as adulterant cause:

- A) Cancer
- B) Liver disorder
- C) Stomach pain
- D) Diarrhea

**Answer: B**

62. Turmeric powder adulterated with chemicals may cause:

- A) Liver disorder
- B) Diarrhea
- C) Cancer (carcinogenic effect)
- D) Stomach disorder

**Answer: C**

63. Papaya seeds in pepper cause:

- A) Diarrhea
- B) Liver disorder
- C) Cancer
- D) Stomach disorder

**Answer: A**

64. Brown sugar and water in honey cause:

- A) Stomach disorder
- B) Cancer
- C) Liver disorder
- D) Diarrhea

**Answer: A**

65. Tamarind seed powder in coffee causes:

- A) Liver disorder
- B) Stomach disorder
- C) Diarrhea
- D) Cancer

**Answer: C**

66. Red brick powder in chilli causes:

- A) Headache
- B) Cancer
- C) Stomach pain
- D) None

**Answer: C**

67. Which is *not* a way to identify adulteration?

- A) Checking quality
- B) Ensuring clean packing
- C) Looking for harmful substance
- D) Adding more adulterant

**Answer: D**

68. Which of the following is **not matter**?

- A) Sand
- B) Air
- C) Music
- D) Stone

**Answer: C**

69. Matter is made up of:

- A) Molecules only
- B) Tiny particles
- C) Big chunks
- D) Energy

**Answer: B**

70. The force of attraction in gases is:

- A) Strong
- B) Medium
- C) Very weak
- D) None

**Answer: C**

71. Which state of matter can flow easily?

- A) Solid
- B) Liquid
- C) Gas
- D) Both B and C

**Answer: D**

72. Example of something that occupies space but we cannot see:

- A) Stone
- B) Water
- C) Air
- D) Chair

**Answer: C**

73. The SI unit of mass is:

- A) Gram
- B) Kilogram
- C) Pound
- D) Liter

**Answer: B**

74. A substance formed by chemical combination of elements is called:

- A) Mixture
- B) Compound
- C) Alloy
- D) Solution

**Answer: B**

75. The smallest unit of a compound is:

- A) Atom
- B) Molecule
- C) Element
- D) Proton

**Answer: B**

76. Which state of matter has the highest density?

- A) Solid
- B) Liquid
- C) Gas
- D) All same

**Answer: A**

77. Which of the following has no definite shape but definite volume?

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

**Answer: B**

78. Which of these takes the shape of the container?

- A) Solid
- B) Liquid and Gas
- C) Gas only
- D) Solid and Liquid

**Answer: B**

79. Which state of matter can be easily compressed into smaller volume?

- A) Solid
- B) Liquid
- C) Gas
- D) None

**Answer: C**

80. The movement of particles in solids is:

- A) Free movement
- B) Very fast
- C) Vibrations in fixed positions
- D) Random collisions

**Answer: C**

81. The conversion of water into vapor is:

- A) Freezing
- B) Melting
- C) Evaporation
- D) Condensation

**Answer: C**

82. Conversion of vapor into water is:

- A) Melting
- B) Condensation
- C) Evaporation
- D) Sublimation

**Answer: B**

83. The process of separating salt from seawater involves:

- A) Filtration
- B) Evaporation
- C) Winnowing
- D) Threshing

**Answer: B**

84. Freezing means conversion of:

- A) Liquid to solid
- B) Solid to liquid
- C) Gas to liquid
- D) Liquid to gas

**Answer: A**

85. The reverse of freezing is:

- A) Evaporation
- B) Melting
- C) Condensation
- D) Sublimation

**Answer: B**

86. The process of changing directly from solid to gas is:

- A) Condensation
- B) Sublimation
- C) Evaporation
- D) Liquefaction

**Answer: B**

87. Threshing is used in:

- A) Milk and cream separation
- B) Separating husk from grains
- C) Separating grains from stalks
- D) Removing salt from water

**Answer: C**

87. To separate sand and salt dissolved in water, we need:

- A) Only filtration
- B) Only evaporation
- C) Both filtration and evaporation
- D) Winnowing

**Answer: C**

88. Butter from milk is separated by:

- A) Decantation
- B) Sedimentation
- C) Churning
- D) Sieving

**Answer: C**

89. Which method uses a magnet?

- A) Winnowing
- B) Filtration
- C) Magnetic separation
- D) Decantation

**Answer: C**

90. An example of winnowing is:

- A) Tea leaves from tea
- B) Husk from wheat
- C) Salt from water
- D) Iron from sand

**Answer: B**

91. Which method is used in oil extraction from cream?

- A) Churning
- B) Winnowing
- C) Sieving
- D) Filtration

**Answer: A**

92. The clear liquid after sedimentation is called:

- A) Residue
- B) Filtrate
- C) Supernatant liquid
- D) Solute

**Answer: C**

93. Filtration separates:

- A) Soluble solids from liquids
- B) Insoluble solids from liquids
- C) Two solids
- D) Two gases

**Answer: B**

95. Brass is a mixture of:

- A) Copper and Tin
- B) Copper and Zinc
- C) Zinc and Iron
- D) Copper and Aluminium

**Answer: B**

96. The mixture of alcohol and iodine is called:

- A) Tincture of iodine
- B) Salt solution
- C) Alloy
- D) Mineral

**Answer: A**

97. Aerated drink is a mixture of:

- A) Only water
- B) Water and sweetener
- C) Water, CO<sub>2</sub>, sweetener, and colouring agent
- D) Only CO<sub>2</sub> and water

**Answer: C**

98. The percentage of gold in 22k gold is:

- A) 75%
- B) 91.6%
- C) 99.9%
- D) 80%

**Answer: B**

99. Which of the following is a **homogeneous mixture**?

- A) Air
- B) Sand and salt
- C) Rice and stones
- D) Wheat and husk

**Answer: A**

100. Which of the following is a **heterogeneous mixture**?

- A) Salt solution
- B) Air
- C) Oil and water
- D) Brass

**Answer: C**